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Functional testing 60m



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There are two test types: functional and nonfunctional testing.

Functional testing

Functional testing checks and evaluates how well a system (or part of a system) or app performs its functions against what is required by the specifications documents, including the business requirements, technical specifications, user stories or functional specifications. Sometimes the functionality may be undocumented, but functional testing should test what the system should do despite this.

Functional testing concentrates on the following:

- **Main functionality of the system**
- **Basic usability:** Can a user easily navigate the system or application?
- **Accessibility:** How accessible is the system navigation or interface for people with impairments?
- **Error conditions:** Are suitable error messages conveyed in error conditions?

Examples of functional tests include testing whether:

- a user can successfully log in to a web application when they provide valid credentials
- the payment system displays an error message and refuses to proceed with completing the order if a user enters an incorrect credit card number
- when a user fills out a registration form, their records are successfully added to the database.

Functional testing simulates actual system usage, making no assumptions about system structure or underlying processing. In this way, it makes use of black box testing techniques (which we will cover shortly). Testing functionality can be done via:

- **requirement-based testing**, which takes the specifications for all the functional requirements of an application and uses these to form the basis of the tests
- **business process-based testing**, which uses knowledge of the business processes that describe the scenarios involved in the day-to-day usage of the system in a business.

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How to perform functional testing

Both approaches to functional testing – requirement-based and business process-based – are completed via a control flow path, which is outlined below.

1. Identify and provide appropriate test input data
2. Compute the expected outcomes with the test input values
3. Execute test cases
4. Compare the actual and computed expected results.

When performing a functional test case, it is useful to use a template to record the test data, steps and expected results. The following is an example of a test case that documents the functional testing of a login page.

Test case ID	TC_01		
Test case description	Verify whether a user is able to log in with a valid username and password		
Test data	Username: JackandJill Password: 123456		
Test steps	Expected test result	Actual test result	Status pass/fail
1. Open URL: {website homepageurl} 2. Click on Login button from header and navigate to login page 3. Enter registered user's username in the username field 4. Enter valid password 5. Click on Login button	User should be logged in successfully	Same as expected	Pass

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Test case ID	TC_02		
Test case description	Verify whether the app allows a user to log in with a valid username and an invalid password		
Test data	Username: JackandJill Password: 1233333		
Test steps	Expected test result	Actual test result	Status pass/fail
1. Open URL: {website homepageurl} 2. Click on Login button from header and navigate to login page 3. Enter registered user's username in the username field 4. Enter invalid/incorrect password 5. Click on Login button	1. It should not allow a user to log in 2. It should display a validation message for invalid credentials 3. Incorrect input field should get highlighted in a different colour	Same as expected	Pass

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Test case ID	TC_03		
Test case description	Verify the login page when the fields are blank and Login button is clicked		
Test data	N/A		
Test steps	Expected test result	Actual test result	Status pass/fail
1. Open URL: {website homepageurl} 2. Click on Login button from header and navigate to login page 3. Click on Login button	1. It should not allow a user to log in 2. Validation message should be displayed to enter credentials 3. Username and password fields should get highlighted in a different colour	Same as expected	Pass



In the following activity, you are required to practise writing test cases for the functional testing of a login page on a web application.

The test cases you create for your functional testing should test the following situations:

1. User clicks the Forgot password button, and an email is sent.
2. Create a new password using valid data from the Forgot password email.
3. Log in to the account using the new password.

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Note: You are not required to perform actual testing based on these situations. The purpose of this activity is for you to practise writing the test cases and to indicate what type of test data you'd use, the steps you'd take and the results you'd expect for each situation. We will practise performing the actual test later in this topic.

The following pre-conditions apply to the situations above:

- The user must have a valid email ID and be registered with the system
- The user must be an active user.
- The example test cases below demonstrate these pre-conditions:

Test case ID	Test case description	Test data	Test steps	Expected test results
TC-07	1. Verify the email text field when it has an email address without the @ symbol. 2. Verify the email text field when it has a random string of characters instead of a real email. 3. Verify the email text field that has the @ symbol written in words. 4. Verify the email text field that has a missing dot in the email address	testAtmail9.io test@mail9io test@mail9 @mail9.io	1. Go to the site. 2. Click the button to create an account. 3. Enter the invalid email addresses. 4. Click on the Register button.	The system should show a validation error message for invalid email addresses.

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TC-08	1. Verify all valid email addresses	test.45@mail7.io test@mail7.io	1. Go to the site. 2. Click the button to create an account. 3. Enter valid email addresses. 4. Click on the Register button.	The system should not show any validation message.
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☒ Test case 1

Non-Assessed

Write a test case for the following scenario:

1. User clicks the Forgot password button, and an email is sent.

You may want to use this [template](#) to create your testplan. Upload your plan once complete.

Add file

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☒ Test case 2

Non-Assessed

Write a test case for the following scenario:

1. Create a new password using valid data from the Forgot password email..

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[Add file](#)**Submit**☒ Test case 3

Non-Assessed

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Write a test case for the following scenario:

- Log in to the account using the new password.

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Thoughts on this page?

